

Grade 12

17 Feb

## Question 1

1.1.  $m=60$  ;  $n=84$  (2)

1.2.

12 ; 24 ; 40 ;

12      16

4

$$2a = 4 \quad \checkmark$$

$$a = 2$$

$$3a + b = 12 \quad \checkmark$$

$$3(2) + b = 12$$

$$\therefore b = 6$$

$$a+b+c=12 \checkmark$$

$$2 + 6 + c = 12$$

$$C = 4$$

$$\therefore T_n = 2n^2 + 6n + 4 \quad (3)$$

1.3.  $2n^2 + 6n + 4 = 312$  ✓  $T_n = 312$

$$2n^2 + 6n - 308 = 0$$

$$n^2 + 3n - 154 = 0 \quad \checkmark \text{ SF}$$

$$(n - 11)(n + 14) = 0$$

$$n = 11$$

or

$$n = -14$$

N/A ✓ reject

$\therefore$  11<sup>th</sup> Day ✓

(4)

## Question 2

2.1.  $T_2 - T_1 = T_3 - T_2$  ✓m

$$(p+5) - (2p-3) = (2p+7) - (p+5) \quad \checkmark A$$

$$p+5 - 2p+3 = 2p+7 - p-5$$

$$-2p = -6$$

$$\therefore p = 3 \quad \checkmark$$

(3)

2.2.  $3 + 8 + 13 + \dots + 173$

$$T_n = a + (n-1)d$$

$$= 3 + (n-1)5$$

$$= 3 + 5n - 5$$

$$T_n = 5n - 2 \quad \checkmark T_n$$

$$5n - 2 = 173 \quad \checkmark T_n = 173$$

$$5n = 175$$

$$n = 35 \quad \checkmark$$

(3)

$$2.3. \quad S_n = \frac{n}{2} [2a + (n-1)d] \quad \checkmark \text{Using correct formula}$$

$$S_{35} = \frac{35}{2} [2(3) + (34)5] \quad \checkmark^{CA} n, a, d$$

$$= 3080 \checkmark \quad (3)$$

$$2.4. \quad 8 + 18 + 28 + \dots + 168 \quad \checkmark \text{even}$$

$$S_{17} = \frac{17}{2} [2(8) + (16)10] \quad \checkmark \text{Correct Formula}$$

$\checkmark a, d$

$\checkmark n=17$

$$= 1496 \checkmark$$

(5)

Alternate:

$$3 + 13 + 23 + \dots + 173 \quad \checkmark \text{odd}$$

$$S_{18} = \frac{18}{2} [2(3) + 17(10)] \quad \checkmark \text{Correct formula}$$

$a, d$

$$= 1584$$

$\checkmark n$

$$\therefore 3080 - 1584$$

$$= 1496 \checkmark$$

(5)

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[14]

### Question 3

3.1.1.  $T_n = ar^{n-1}$

$$T_n = 27 \left( \frac{1}{3} \right)^{n-1}$$

✓ Correct formula; a  
✓ r

$$= 3^3 \cdot (3^{-1})^{n-1}$$

$$= 3^{-n+4}$$

✓ Single base 3.

(3)

3.1.2.  $3^{-n+4} = \frac{1}{2187}$

✓  $T_n = \frac{1}{2187}$

$$3^{-n+4} = 3^{-7} \quad \checkmark$$

$$\therefore -n+4 = -7$$

$$\therefore n = 11 \quad \checkmark$$

(3)

1. 3.  $S_n = \frac{a(1-r^n)}{1-r}$

$$S_7 = \frac{27 \left( 1 - \left( \frac{1}{3} \right)^7 \right)}{1 - \frac{1}{3}}$$

✓ Correct formula  
subs

$$= \frac{1093}{27}$$

✓ Either answer  
accepted.

$$\approx 40,48$$

(2)

$$3.2.1. \quad r = \frac{1-x}{5} \quad \checkmark r$$

$$-1 < r < 1$$

$$-1 < \frac{1-x}{5} < 1 \quad \checkmark -1 < r < 1$$

$$-5 < 1-x < 5$$

$$-6 < -x < 4$$

$\checkmark$  Answer

$$6 > x > -4 \quad ; \quad -4 < x < 6 \quad (3)$$

$$3.2.2. \quad a = \frac{3(1-(-1))}{5} = \frac{6}{5} \quad r = \frac{2}{5}$$

$\checkmark a; r$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{6}{5} \div \left(1 - \frac{2}{5}\right) \quad \checkmark \text{subs in correct formula}$$

$$= 2 \quad \checkmark \quad (3)$$

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[14]

### Question 4

$$4.1. T_8 + T_{10} = 28$$

$$S_{12} = 258$$

$$a + 7d + a + 9d = 28$$

$$\frac{12}{2} [2a + 11d] = 258$$

$$2a + 16d = 28 \dots \textcircled{1} \checkmark$$

$$\therefore 2a + 11d = 43 \dots \textcircled{2} \checkmark$$

$$\textcircled{1} - \textcircled{2}$$

$$5d = -15 \quad \checkmark m$$

$$d = -3 \quad \checkmark$$

$$\therefore 2a + 16(-3) = 28 \quad \checkmark m$$

$$2a = 76$$

$$a = 38 \quad \checkmark$$

(6)

$$4.2. \sum_{k=3}^3 8(2)^{k-1} > 5500$$

$$32 + 64 + 128 + \dots \checkmark \quad r=2$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

✓ a & r into  
correct formula

$$\checkmark S_n \quad 5500 = \frac{32(2^n - 1)}{2 - 1}$$

$$2^n = \frac{1383}{8} \quad \checkmark$$

$$n = \log_2 \left( \frac{1383}{8} \right)$$

$$n \approx 7.43 \quad \checkmark \text{ Use log.}$$

$$\therefore n = 8 \quad \checkmark \text{ rounded up}$$

$$m - 3 + 1 = 8$$

$$m = 10 \quad \checkmark \quad n = \text{top} - \text{bottom} + 1 \quad (7)$$

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